



# Stacking segment-based CNN with SVM for recognition of atrial fibrillation from single-lead ECG recordings

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## ABSTRACT

**Background and objective:** Atrial fibrillation (AF) is the most common form of cardiac rhythm disorder. Early detection of AF can result in a lower risk of stroke, heart failure, systemic thromboembolism, and coronary artery disease. AF detection however is challenging due to the need for specialised equipment and professional technicians. Hand-held electrocardiogram (ECG) devices, including wearables, are now available and provide a potential mechanism for detecting AF. We wished to identify AF from short single-lead ECG recordings using a machine learning method.

**Methods:** We predicted AF from ECG signals by stacking a support vector machine (SVM) on statistical features of segment-based recognition units produced by a convolutional neural network. We used the ECG dataset from the PhysioNet/Computing in Cardiology Challenge 2017, which contained 8528 ECG recordings, to validate our method.

**Results:** ECG recordings were categorised into four classes with an average F1 score of 84.19% under fivefold cross-validations.

**Conclusions:** Our model performed better than other state-of-the-art methods applied to the same dataset using the same metric. This stacking method can be generalised for other problems related to medical signals as it does not require expertise in analysing ECG data.

## 1. Introduction

Atrial fibrillation (AF) is one of the most common heart arrhythmias, and involves a fast and irregular heart rhythm [1]. It is a supraventricular tachyarrhythmia characterised by uncoordinated atrial activation with deterioration of atrial mechanical function [2]. AF is a major public health problem occurring in 1–2% of the population. The prevalence of AF is higher in men than in women and increases with age, with highest incidence amongst those over 75 years old [3]. AF prevalence and incidence are increasing because the global population age is increasing [4] and survival from conditions predisposing to AF (e.g., coronary heart disease) is improving [5]. The life-time risk of developing AF by 55 years

of age is 24.8% for men and 22.9% for women [6]. For those people with AF, there is a heightened risk of more severe cardiovascular conditions, such as stroke and heart failure [7]. Importantly, patients may have a completely asymptomatic arrhythmia or occasional episodic symptoms of AF which can remain undetected [8]. Often the first presentation with AF is associated with an acute AF-related complication including stroke, heart failure, systemic thromboembolism, and coronary artery disease. Early detection of AF is therefore needed to enable early proactive treatment.

The most common method of detecting AF is through the observation and analysis of electrocardiogram (ECG) signals [9]. The ECG is a simple test that measures and records the electrical potential of the heartbeat.

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These signals are used to determine the presence of heart-related problems. While specialised equipment usually provides multi-channel signals, a single-lead ECG recording from hand-held ECG devices (including wearables) has only one channel to represent the variations of the electrical signal generated by the heart during contraction. Measuring an ECG signal can be resource intensive due to the need for specialised equipment and professional technician support. Hand-held or wearable ECG devices have been significantly improved [10]. Non-practitioners can record their own signals to enable early detection of AF. AF can be detected even if symptoms are episodic, because self-monitoring can take place over longer periods. However, problems are often encountered in the interpretation of ECG signals. This includes ECG recordings having short lead times, with signals being prone to artifact due to instrumentation manipulation by non-technical users. The use of algorithms to improve the automatic analysis and diagnoses of AF using self-monitored ECG signals is now needed.

Previous research on AF detection using ECG signals has focused on extracting hand-engineered features, such as heart rate-based, P-wave, T-wave, and frequency-domain features. This has required expertise in the reading of ECG signals and heart arrhythmias [11–19]. Steps to recognise AF using ECG signals often employs well-known classifiers, such as Random Forests (RF) [20] or AdaBoost [21]. In 2017, PhysioNet, a well-known platform hosting an annual series of challenges focusing on unsolved problems in clinical and basic science, organised a challenge to detect AF from short single lead ECG recordings [22]. This was a challenging problem due to the broad taxonomy of rhythms. Also, many non-AF rhythms had irregular RR intervals (the time elapsed between two successive R-waves) which could be mistaken for AF. Three out of the four teams who shared first place in the challenge (with  $F_1$  scores from 0.825 to 0.831) exploited hand-engineered features [23–25]. Zabihi et al. [24] proposed a method containing three steps, which were similar to Sopic et al. [26]: preprocessing, feature extraction and classification. Their method extracted 491 features from multiple domains, including time, frequency, time-frequency, phase space, and meta-level. An RF classifier was dedicated to select 150 of the most important features for classification by another RF model.

The method with the highest  $F_1$  score in the PhysioNet 2017 challenge was from Teijeiro et al. [23]. Their method utilised rhythm, morphological, and signal quality features for a two-step classification. Features were first processed by two independent classifiers for global and sequence classification. Then a stacking linear discriminant analysis model combined global and sequence classification to yield class predictions.

Deep learning (DL) has been used in a wide variety of fields as this method can be used to extract useful deep features without expert knowledge of the dataset. Many researchers have also applied DL on ECG signal classification [9,27–32]. Pyakillya et al. utilised a Convolutional Neural Network (CNN) of seven convolutional layers and three fully-connected layers to process the raw ECG signals for signal classification [30]. Yao et al. computed the hand-crafted features, instant heart rate from RR intervals, then used a multi-scale CNN (MCNN) for classification [28]. The MCNN outperformed conventional methods in detecting AF according to ROC curves and AUC measurements. Sujadevi et al. compared many architectures of Recurrent Neural Networks (RNNs), LSTMs [33], and GRUs [34] for processing raw ECG signals [35]. In their experiments, LSTM and GRU with 64 memory blocks were shown to perform best.

Ng's group proposed a CNN for classifying various classes of single-lead ECG signals [36,32]. Their DL model exceeded the performance of board-certified cardiologists in detecting a wide range of heart arrhythmias. They used a larger dataset than previously and trained a very deep CNN of 33 convolutional layers and 16 residual blocks. Their model achieved a  $F_1$  score of 0.83 on a hidden testing set.

In the PhysioNet 2017 challenge, in addition to CNN and CRNN [37], DL was widely applied to extract deep features and AF classification by other top teams [37–39]. Also ranked fifth with CRNN, Xiong et al. used

a 16-layer CNN with their proposed skip connections to predict fixed-length segments of raw ECG data [38]. Final predictions were classes with the highest votes by the CNN on segments. In another paper, Hong et al. passed three sources of features, comprising expert, center-wave, and deep features, to an ensembled XGBoost [40]. Deep features were produced by a deep residual network processing sampled segments and an LSTM with centerwave features as inputs. This method achieved first place with three other teams in the challenge.

The limitations of the above studies are the lack of dataset balancing, final predictions by using simple voting or the use of hand-crafted features. Although hand-engineered features are good indicators for detecting AF, they theoretically can have bias. In this study, we used a publicly available and highly imbalanced dataset from the PhysioNet 2017 challenge [22] to propose a novel pipeline for classifying ECG signals. Our method used segment-based recognition units with different segment shifts to address the imbalance issue in data and support signals which are varied in length. We developed a CNN architecture to automatically classify each segment without the need of using feature engineering. Our CNN was trained using an adaptive learning rate changing strategy without using any validation data to make use of the whole small training set while avoiding over-fitting. We also proposed stacking a support vector machine (SVM) with outputs from CNN for final classification of each ECG signal. This has been shown to produce a better performance than the more common hard-voting method. In the following sections, we describe our novel methodological approaches using the PhysioNet 2017 dataset.

## 2. Material and methods

### 2.1. Dataset and balancing data

The data input of the 8528 instances from the PhysioNet 2017 dataset was used. These ECG signals had lengths ranging from 2200 to 18000 data points (from 7.3 to 60 s) recorded at 300 Hz. Four possible classes for a recording included *Normal* ( $n$ ), *AF* ( $a$ ), *Others* ( $o$ ), and *Noise* ( $p$ ). The number of recordings in the four classes were 5076, 758, 2415, and 279, respectively. The classification performance defined in the challenge was the  $F_1$  score, which is the average class- $F_1$  score across the classes  $n$ ,  $a$ , and  $o$ :

$$F_1 = \frac{1}{3} \sum_{i \in \{n,a,o\}} F_{1i}, \quad (1)$$

where  $F_{1n} = \frac{2 \times Nn}{\Sigma_N + \Sigma_n}$ ,  $F_{1a} = \frac{2 \times Aa}{\Sigma_A + \Sigma_a}$ ,  $F_{1o} = \frac{2 \times Oo}{\Sigma_O + \Sigma_o}$ , and the counting rules for the numbers of the metrics are defined in Table 1.

We employed fivefold cross-validation with stratified 1707, 1706, 1706, 1705, and 1704 recordings in 5 folds to estimate the  $F_1$  score using our method. Each ECG recording was divided into fixed-length segments of 4096 points. The difference between the starting points of any two successive segments on a recording is constant for each class and called the (segment) shift. These segments were fed to the stacking method to predict a class for each recording (explained in Section 2.2). The first step of the stacking method is the segment-based prediction by a convolutional neural network (CNN). We used these fixed-length segments

**Table 1**  
Definition of parameters for scoring used in Eq. (1).

|        | Predicted classification |            |            |            | Total      |
|--------|--------------------------|------------|------------|------------|------------|
|        | Normal                   | AF         | Others     | Noise      |            |
| Normal | Nn                       | Na         | No         | Np         | $\Sigma_N$ |
| AF     | An                       | Aa         | Ao         | Ap         | $\Sigma_A$ |
| Others | On                       | Oa         | Oo         | Op         | $\Sigma_O$ |
| Noise  | Pn                       | Pa         | Po         | Pp         | $\Sigma_P$ |
| Total  | $\Sigma_n$               | $\Sigma_a$ | $\Sigma_o$ | $\Sigma_p$ |            |

to train the CNN. Since the dataset was highly imbalanced, we sampled with different shifts between input segments for each class to equalise the distribution of the four classes. The segment shifts were manually determined based on the following relationship for one recording:

$$ss_i = \frac{l_t - l_s}{ns_i - 1} \text{ because,} \quad (2)$$

$$ns_i = \frac{l_t - l_s}{ss_i} + 1, \quad (3)$$

where  $ns_i$  is number of segments of an ECG recording of class  $i$ ,  $l_t$  is the length of the recording,  $l_s$  is segment length (fixed to 4096), and  $ss_i$  is the segment shift of class  $i$ . In the training dataset, the segment shift  $ss_i$  of class  $i$  was chosen to result in a number of segments  $ns_i$  within the range of (20,000, 22,000). Hence, each class will have a different segment shift to balance the numbers of instances in all classes. Therefore, according to Eq. (2), four segment shifts in the training data were manually refined as follows: (Normal, AF, Others, Noise) = (1142, 150, 560, 35).

In testing, the segment shift was constant as the target class was unknown. Hence, the distribution of classes in the test set was unchanged. The constant segment shift of 1 s, which was equal to the shift used by Rajpurkar et al. [36], was used in our experiment for testing. This shift is equivalent to 300 sampling points at the recording frequency of 300 Hz.

## 2.2. Stacking SVM on segment-based CNN predictions

Given the signal segments, we used a stacking method to process the class prediction of an ECG recording with two main steps as depicted in Fig. 1. The first step was to use a deep CNN to provide the initial class predictions of these fixed-length segments. Because an ECG recording is generally comprised of multiple temporal segments, each recording received a sequence of predictions which were generally not identical. Thus, a second step to determine the final hypotheses of ECG recordings was involved. We used a four-class SVM model as a stacking classifier on the statistical features of the prediction sequences of ECG signals. For a recording, classes were counted and normalised from its sequence of CNN predictions to form a 4-component vector  $\hat{\theta}_i$  of normalised class-counts corresponding to four classes as follows:

$$\hat{\theta}_i = \frac{\theta_i}{\sum_{j \in \text{all 4 classes}} \theta_j}, \quad (4)$$

where  $\theta_i$  is the number of segment-based predictions of class  $i$ . Each ECG recording was therefore assigned a vector of class-counts summing to 1. This vector was passed to the SVM model to predict the class of the recording. The training processes of the CNN and the SVM model were done sequentially.

We utilised the stacking SVM model to account for issues of episodic AF in the class AF as well as unstable signals in other classes. For instance, a recording can accommodate only a few segments which shows the unique characteristics of its class. Consequently, the class with the highest vote may not refer to the actual class of a signal. We employed a radial basis function (RBF) kernel for the SVM model to allow nonlinearities in decision boundaries between classes as this generally improves SVM performance. The RBF kernel was optimised using a grid search with cross-validation on the training set of the first split of the fivefold cross-validation.

## 2.3. Deep neural network architecture

We used the architecture shown in Fig. 2 for our CNN to process directly fix-length segments of ECG signals. The architecture started with 4096 input neurons corresponding to the length of segments. The hidden layers of the feature extraction part have six 1-D convolutional layers (kernel size = 15, padding = 7, and stride = 1) with which each is followed by batch normalisation (BN) and a max-pooling layer (kernel size = 4). The number of feature maps increased from the first convolution layer with 64 feature map to the last one with 256 deep features. The classification part has two hidden fully connected (FC) layers of size 1024 and 512 respectively, and an output layer of four neurons for four classes. All hidden layers had rectified linear unit (ReLU) activations and dropouts of probability 0.5.

## 2.4. CNN training strategy with a decreasing learning rate

We proposed a strategy to train the CNN in Algorithm 1. This strategy can result in faster training with large learning rates in the beginning and small learning rates in the final phase of training. Large learning rates can optimise the weights of the CNN faster and help escape saddle points or local optima in the early phase. Small learning rates in the later phase can fine-tune the CNN in small regions of optimal points.

A transition to the next smaller learning rate was activated when either reaching ten epochs with one learning rate to avoid overfitting or two consecutive epochs of slowly decreasing training loss ( $p < 2\%$ ). The latter criterion, slow or no decrease of training loss, was driven by the fact that it can be a sign of too large learning rates, which led to fluctuations around an optimal point [41]. Instead of using a validation set for early stopping, the training process stopped when there were no more learning rates to switch to. This strategy was to maximise the number of training instances. The final model for testing was the one which had the smallest loss in the whole training process.

**Algorithm 1.** CNN training strategy. Final model is saved in the target file

```

1: predefined decreasing learning rates lr_s_teps = [1e-3, 5e-4, 1e-4, 5e-5]
2: best_train_loss = 1000
3: maximum number epochs per step max_epochs = 10
4: with the first learning rate, first training loss loss_pre =
   train_optimizer(lr_s_teps[0])
5: for lr in lr_s_teps do
6:   counter of consecutive epochs with insignificant decrease of training loss
   counter = 0
7:   for epoch in range(max_epochs) do
8:     current loss loss_cur = train_optimizer(lr)

```

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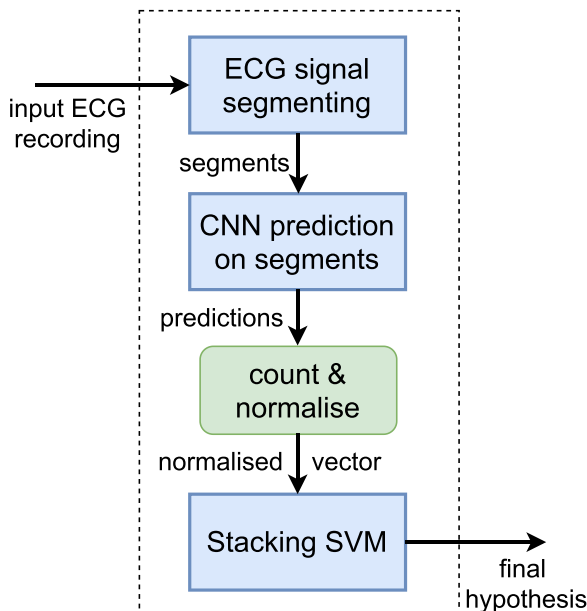


Fig. 1. Main steps of our method to process ECG recordings.

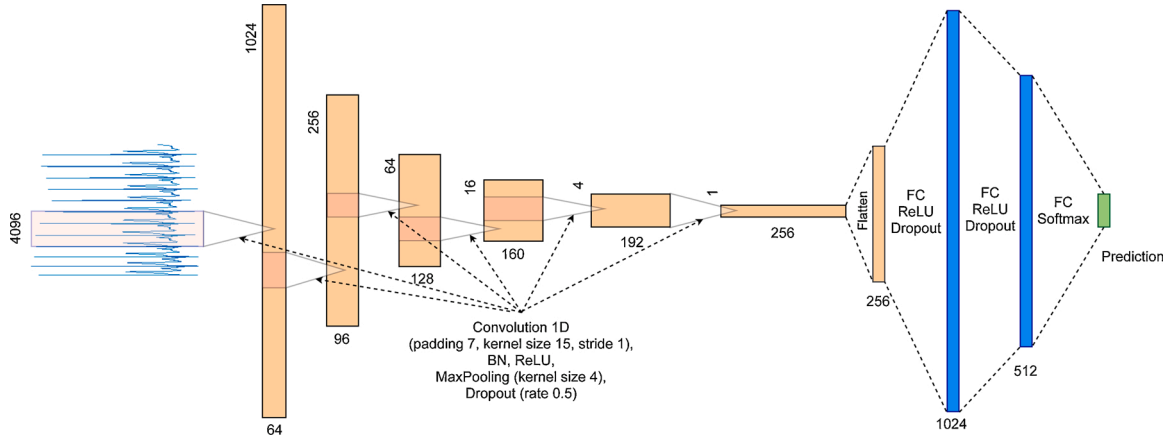


Fig. 2. Our proposed CNN architecture.

(continued)

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9:   if best_train_loss > loss_cur then
10:     best_train_loss = loss_cur
11:     save the current weights to the target file
12:   decreasing rate  $p = (\text{loss}_{\text{pre}} - \text{loss}_{\text{cur}}) / \text{loss}_{\text{pre}}$ 
13:   loss_pre = loss_cur
14:   if insignificant decrease of loss  $p < 0.02$  then
15:     counter = counter + 1
16:   else
17:     reset counter = 0
18:   if counter == 2 then
19:     counter = 0
20:     break to jump to the next smaller learning rate

```

### 3. Results and discussion

We evaluated the proposed method on the publicly available training set of the PhysioNet 2017 dataset (the test dataset is kept hidden for submitted algorithms) using a fivefold stratified cross-validation. For a reliable estimation of the performance of our model, we performed model evaluation in five trials in which each trial differed from the others in the random seed used in stratified sampling. The computing platform was a 3.2 GHz Intel Core i7-8700 desktop computer with 32 GB RAM, 512 GB SSD and an NVIDIA 2080 Ti GPU. On average, for each fold, it took about 10 min to train the model and about 15 s to complete testing. Table 2 shows the detailed  $F_1$  scores and Table 3 shows the precision and recall values of the stacking SVM model in our first trial. The tuned hyperparameters of the RBF kernel in our SVM model were  $\gamma = 1000$  and  $C = 1$ . Table 2 compares the hard voting method, which simply takes the mode of all segment-based CNN predictions of each ECG recording, and the segment-based stacking method using SVM on normalised CNN predictions.  $F_1$  scores of *Normal* signals were generally the highest in both methods. Classifying *Others* was harder using the simpler method, i.e., hard voting, with the lowest  $F_{1o}$  scores. Stacking SVM remarkably upgraded  $F_{1a}$  scores of classes *Normal* and *Others* while maintaining  $F_{1a}$  of class *AF*. The stacking method using SVM with the  $F_1$  score of 0.8368 substantially improved the classification

Table 3

Precision and recall values of the stacking SVM model from a fivefold cross-validation trial (Trial 1).

| Cross-validation | Normal    |        | AF        |        | Others    |        |
|------------------|-----------|--------|-----------|--------|-----------|--------|
|                  | Precision | Recall | Precision | Recall | Precision | Recall |
| Fold 0           | 0.9090    | 0.9537 | 0.8280    | 0.8552 | 0.8160    | 0.7805 |
| Fold 1           | 0.9222    | 0.9349 | 0.7733    | 0.7631 | 0.7718    | 0.8053 |
| Fold 2           | 0.9134    | 0.9566 | 0.8333    | 0.7236 | 0.7923    | 0.8136 |
| Fold 3           | 0.9134    | 0.9566 | 0.8333    | 0.7236 | 0.7923    | 0.8136 |
| Fold 4           | 0.9044    | 0.9418 | 0.8148    | 0.7284 | 0.7650    | 0.7888 |
| Average          | 0.9125    | 0.9487 | 0.8165    | 0.7588 | 0.7875    | 0.8004 |

performance.

Table 4 compares our method with other methods using fivefold cross-validation on the same dataset. Based on this comparison, our method achieved an average  $F_1$  score of 0.8419, which is the highest score among compared methods. When using only DL, the  $F_1$  score 0.8018 of hard voting from Table 2 can be considered competitive compared with other methods. For instance, Zihlmann et al. [37]

Table 4

Results from 5 trials of fivefold cross-validations and comparison with other methods.

| Methods              | $F_1$ score |
|----------------------|-------------|
| Zihlmann et al. [37] | 0.790       |
| CNN                  | 0.792       |
| CRNN                 | 0.8264      |
| Datta et al. [25]    | 0.8368      |
| Stacking CNN-SVM     | 0.8393      |
| Trial 1              | 0.8458      |
| Trial 2              | 0.8425      |
| Trial 3              | 0.8449      |
| Trial 4              | 0.8419      |
| Trial 5              | 0.8419      |
| Average              | 0.8419      |

Table 2

Detailed results of a fivefold cross-validation trial (Trial 1).

| Cross-validation | Hard voting |          |          |        | Stacking SVM |          |          |        |
|------------------|-------------|----------|----------|--------|--------------|----------|----------|--------|
|                  | $F_{1n}$    | $F_{1a}$ | $F_{1o}$ | $F_1$  | $F_{1n}$     | $F_{1a}$ | $F_{1o}$ | $F_1$  |
| Fold 0           | 0.8932      | 0.8082   | 0.6872   | 0.7962 | 0.9308       | 0.8414   | 0.7978   | 0.8567 |
| Fold 1           | 0.9042      | 0.7740   | 0.7247   | 0.8009 | 0.9285       | 0.7682   | 0.7882   | 0.8283 |
| Fold 2           | 0.8939      | 0.8253   | 0.7201   | 0.8131 | 0.9345       | 0.7746   | 0.8028   | 0.8373 |
| Fold 3           | 0.9024      | 0.7894   | 0.7116   | 0.8011 | 0.9354       | 0.7817   | 0.8000   | 0.8390 |
| Fold 4           | 0.8936      | 0.7963   | 0.7028   | 0.7975 | 0.9227       | 0.7962   | 0.7767   | 0.8229 |
| Average          | 0.8975      | 0.7987   | 0.7093   | 0.8018 | 0.9304       | 0.7870   | 0.7931   | 0.8368 |



transformed ECG signals into normalised logarithmic spectrograms before passing them to a CNN or convolutional RNN (CRNN) to output class probabilities. The CRNN method, ranked fifth in the challenge, achieved an F1 score of 0.792 using fivefold cross-validation. Datta et al. [25] classified with cascaded binary classifiers of AdaBoost [21] on hand-engineered features from the literature. This method obtained an F1 score of 0.8264 and was one of the four teams sharing the first place in the challenge.

Our proposed method also has a weakness although this was due to the nature of the dataset. Since there was no information available about where AF or other rhythms occur in each ECG signal, it is possible that a segment from an AF signal may not contain AF. Similarly, a segment from an *Others* signal may not contain any other rhythms. We chose 4096 data points as the length of each segment since this is about half of the whole signal, leading to a high chance that each segment contains important patterns of the signal. In addition, CNNs have the ability to learn local patterns at different positions in a signal. Therefore, our method should be able to achieve a good performance, and in fact, we found that it outperformed the top methods of the PhysioNet 2017 challenge.

#### 4. Conclusions

We have developed and evaluated a segment-based stacking method of CNN and SVM to categorise short single-lead ECG signals into four classes: *Normal*, *AF*, *Others*, and *Noise*. The methodology included segmenting signals into fixed-length segments which also balanced the class distribution, recognising each segment using a deep CNN, and classification on statistical features of segment-based CNN predictions using stacking SVM. The deep CNN with our training strategy successfully extracted useful deep features from raw ECG signals for ECG classification without expertise on ECG signals and cardiac rhythm disorders. By stacking SVM with RBF kernel on the statistics of CNN predictions simultaneously, we were able to fine-tune the decision boundaries between classes. We validated this method on the PhysioNet 2017 dataset and found a superior performance compared with the top teams of the challenge.

Deep learning has been well-known for its ability to extract useful deep features, while SVM can perform well on classifying data with arbitrary numbers inputted. The proposed method could be extended to other problems related to medical signals.

#### Authors' contribution

Quang H. Nguyen: methodology, data curation, software, writing – review & editing. Binh P. Nguyen: conceptualisation, methodology, formal analysis, validation, writing – original draft, writing – review & editing. Trung B. Nguyen: formal analysis, validation, writing – original draft. Trang T.T. Do: visualisation, writing – review & editing. James F. Mbinta: validation, writing – review & editing. Colin R. Simpson: formal analysis, validation, writing – review & editing.

#### Declaration of Competing Interest

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